CPEN 416 / EECE 571O Lecture 10

Quantum compilation I

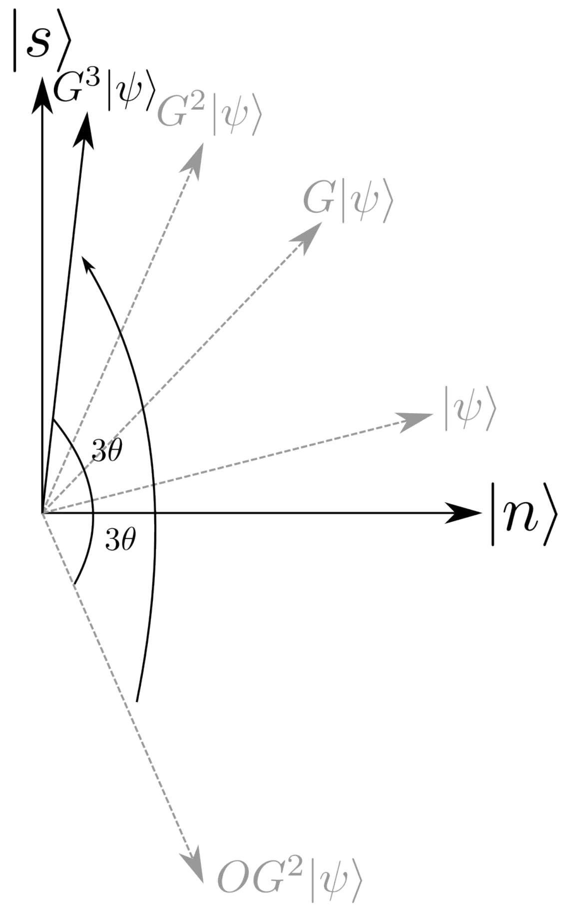
Tuesday 14 October 2025

Announcements

- Quiz 4 today
- Final project group + paper selection **due 17:00 Thursday**

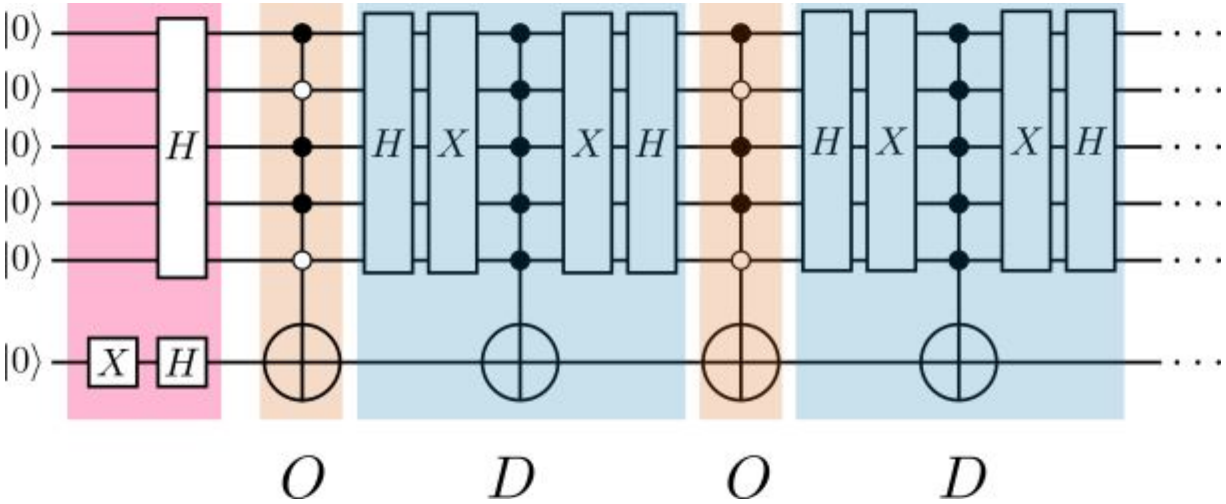
Last time

We discussed how Grover search of an unstructured space has a square-root speedup in **query complexity** over classical methods.



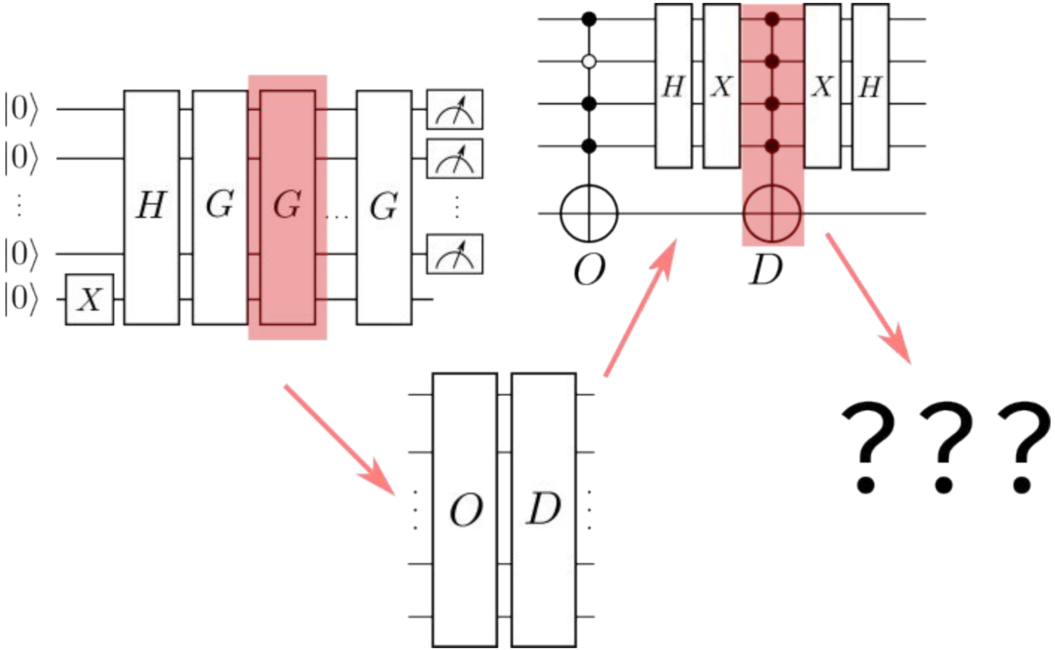
Last time

We implemented Grover's algorithm in PennyLane



Last time

We finished off with this picture



Quiz

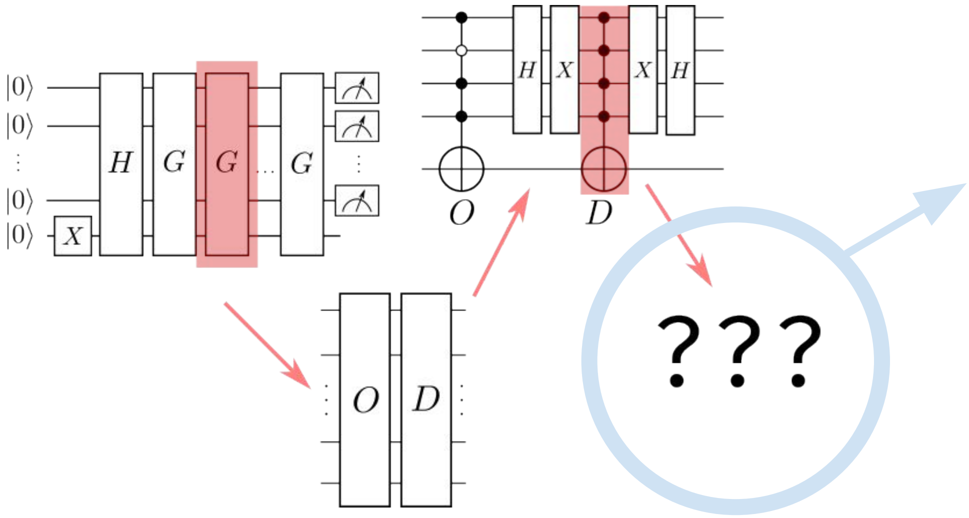
Today

Learning outcomes:

- define and calculate **cost metrics** of quantum circuits
- define and list key **universal** gate sets
- **decompose** selected single- and multi-qubit operations using one- and two-qubit universal gate sets

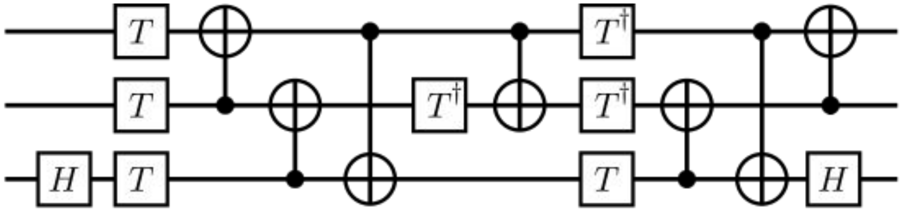
Thursday will involve hands-on component that builds on today’s content (compiling Grover’s algorithm).

Quantum circuit synthesis and optimization



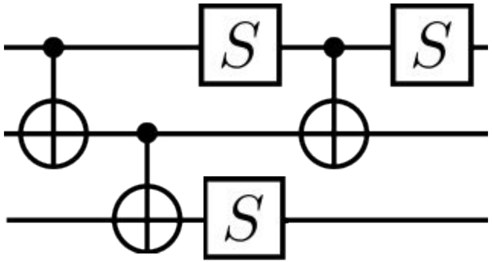
- How do we decide what goes here?
- What criteria determine whether our choice is “good”?
- How do we improve on our choices?

Width, depth, gate count



Circuit depth

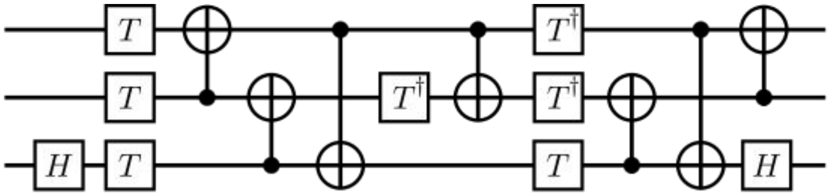
The circuit depth is the length of the longest path in its **directed acyclic graph (DAG)** representation.

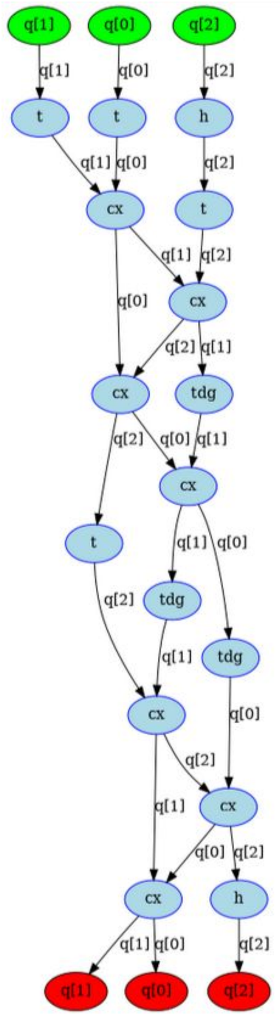


Circuit depth

Generally don't do by hand; NetworkX is your best friend.

https://networkx.org/

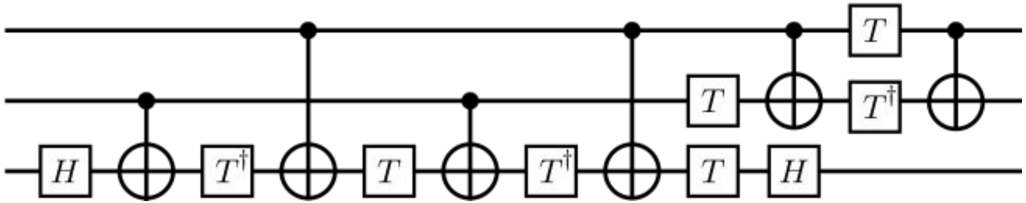




(DAG generated in Qiskit)

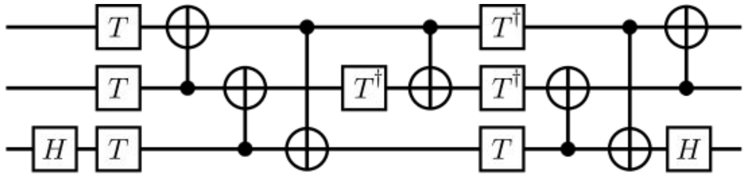
Exercise

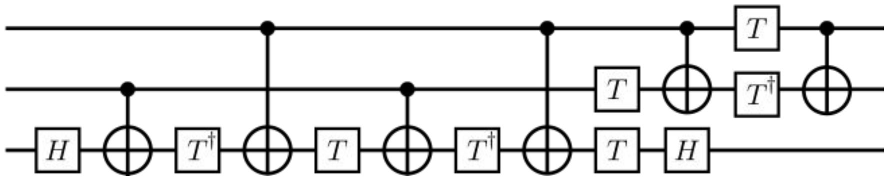
Determine resource counts of the following circuit.



Circuit synthesis and decomposition

What do these circuits do?





Circuit synthesis

Given:

- arbitrary unitary matrix U
- a finite set of gates $\{G_k\}$
- a target precision, ϵ

Find

$$U' = C_1 C_2 \cdots C_m$$

s.t.

$$||U - U'|| < \epsilon, C_i \in \{G_k\}$$

Universal gate sets

A $\{G_k\}$ for which this can be done is called a **universal gate set.**

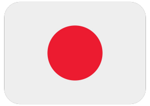
Examples for a single qubit:

- Any two Pauli rotations (exact)
- Hadamard and *T* (approximate)

Theoretical bound: $O(\log^c(1/\epsilon))$ gates, c varies from 1-4 depending on assumptions.

Universal gate sets

Using a different gate set is like *speaking a different language*.

	I would like to use a real quantum computer.	*I would like to use a real quantum computer.*
	J’aimerais utiliser un vrai ordinateur quantique.	*I would like to use a real computer that is quantum.*
	本当の量子コンピュータを作りたい。	*A real quantum computer, I would like to use.*

Universal gate sets

Any single-qubit unitary can be expressed using a sequence of 3 Pauli rotations around a minimum of 2 unique axes

- ZYZ
- XYX
- XYZ
- 9 other combinations *(Euler angles)*

qml.Rot

class **Rot**(*phi, theta, omega, wires, id=None*)

Bases: `pennylane.operation.Operation`

Arbitrary single qubit rotation

$$R(\phi,\theta,\omega) = RZ(\omega)RY(\theta)RZ(\phi) = \begin{bmatrix} e^{-i(\phi+\omega)/2}\cos(\theta/2) & -e^{i(\phi-\omega)/2}\sin(\theta/2) \\ e^{-i(\phi-\omega)/2}\sin(\theta/2) & e^{i(\phi+\omega)/2}\cos(\theta/2) \end{bmatrix}.$$

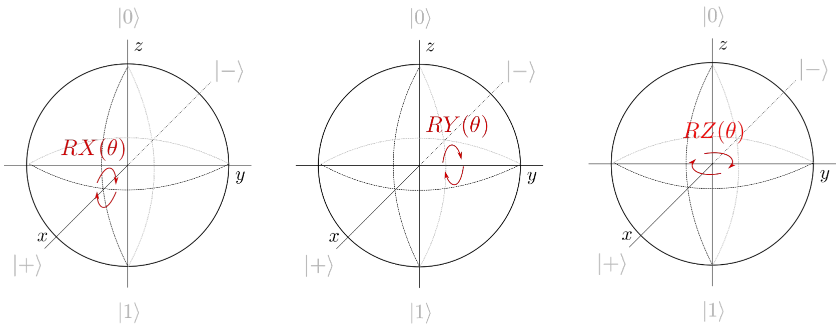
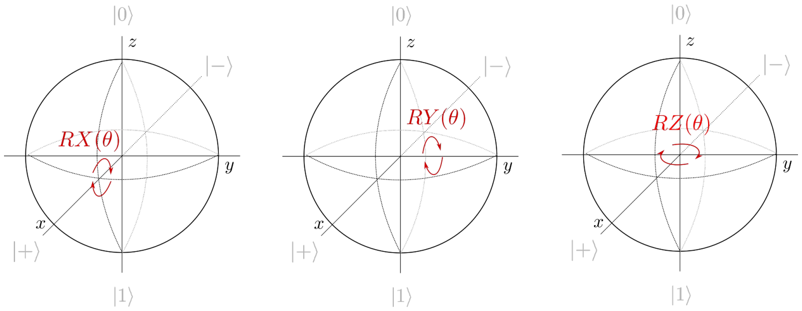


Image: Xanadu Quantum Codebook Node I.6

Hardware-specific example

RX/RY/RZ are **arbitrary-angle** rotations about **fixed axes**



IonQ's trapped ion hardware native gates: *GPI/GPI2*, **fixed-angle** rotations about **arbitrary axes** in *xy*-plane

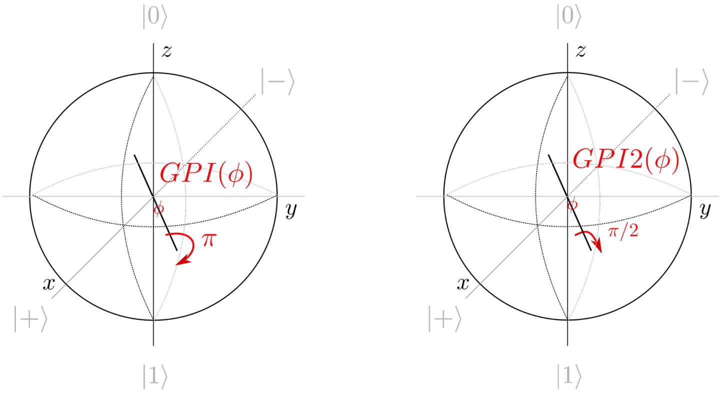
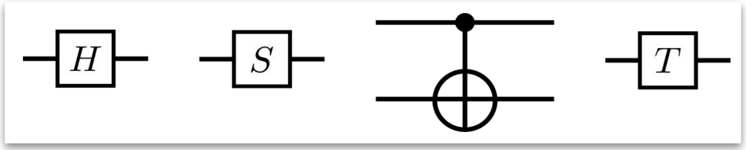
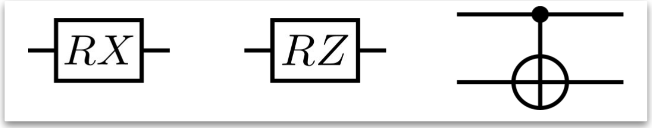


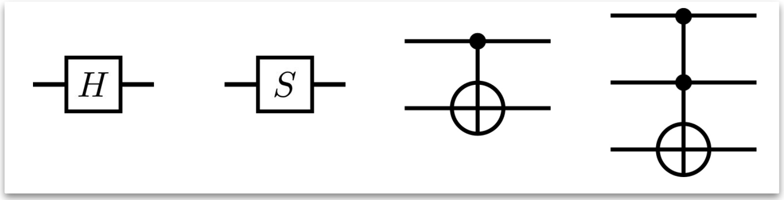
Image adapted from Xanadu Quantum Codebook Node I.6
See blog post: https://pennylane.ai/blog/2023/06/the-ionizer-building-a-hardware-specific-transpiler-using-pennylane

Universal gate sets

Multi-qubit gate sets

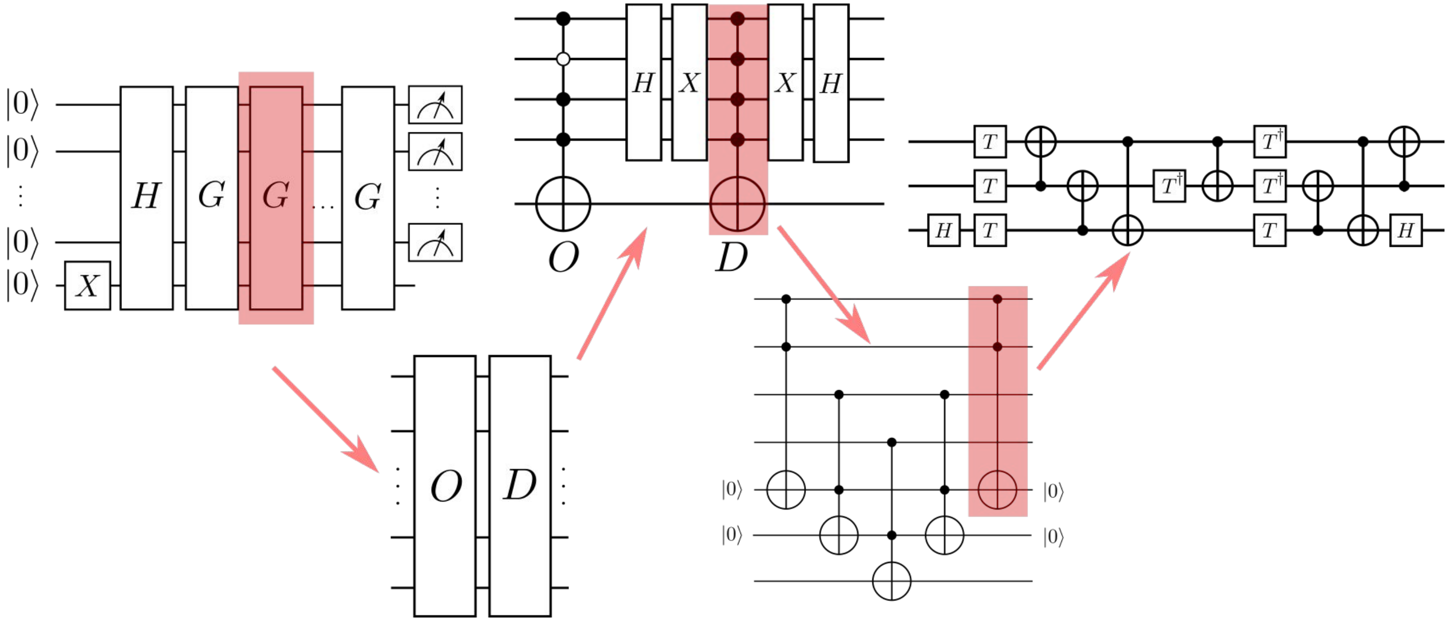






Circuit decomposition

Decompositions of many common operations are **known.** Some are even straightforward to work out by hand.



Exercise

Express Pauli X in terms of H and T.

Hint: start by expressing it using H and Z.

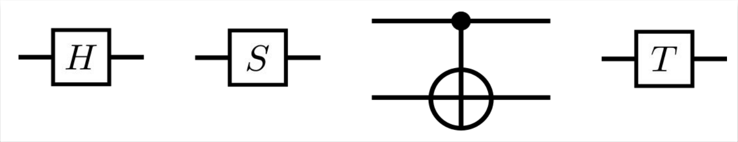
Exercise

Express Hadamard in terms of $\{RX, RY, RZ\}$

Hint: there are many ways to do this!

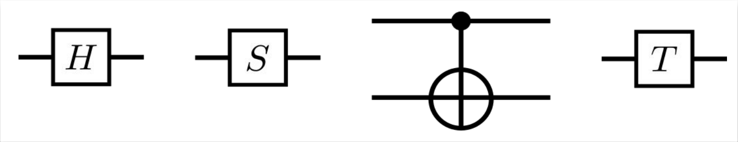
Exercise

Express *CZ* using gates from



Take-home exercise

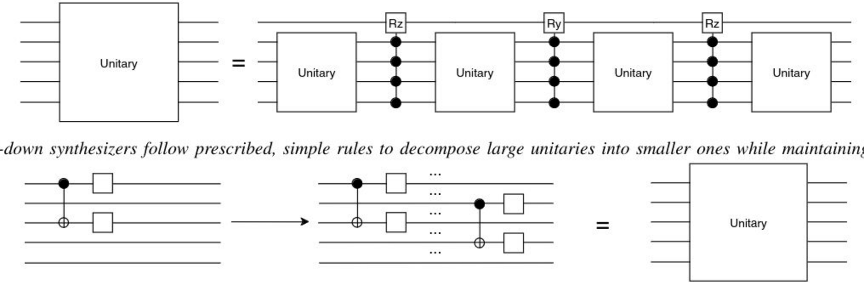
Express controlled Hadamard using gates from



Synthesis and decomposition techniques

For under-specified or arbitrary operations, or for special gate sets, we may need to ***find*** a decomposition.

- search problem
- number-theoretic techniques
- numerical methods



(a) *Top-down synthesizers follow prescribed, simple rules to decompose large unitaries into smaller ones while maintaining equality.*

(b) *Bottom-up synthesizers start with an empty circuit and build up to equality.*

Figure 1: *Quantum synthesizers are either top-down or bottom-up.*

Image: E. Younis, K. Sen, K. Yelick, and C. Iancu. *QFAST: Conflating Search and Numerical Optimization for Scalable Quantum Circuit Synthesis.* arXiv:2103.07093 [quant-ph]

Challenges

Synthesis is hard!

- algorithm complexity and computational resources

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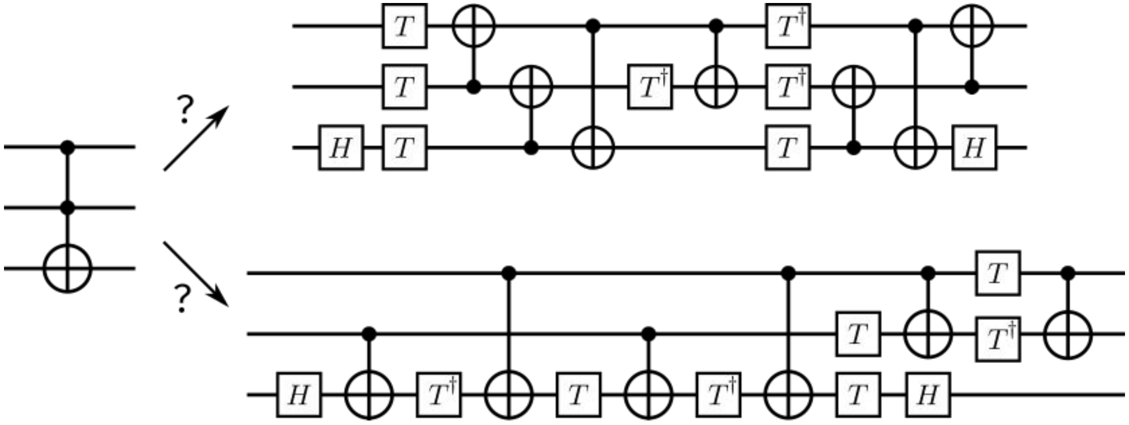
- algorithm complexity and computational resources
- how to choose the best decomposition to *enable further optimization* down the line?



Challenges

Synthesis is hard!

- algorithm complexity and computational resources
- how to choose the best decomposition to *enable further optimization* down the line?
- manage tradeoffs between various resources

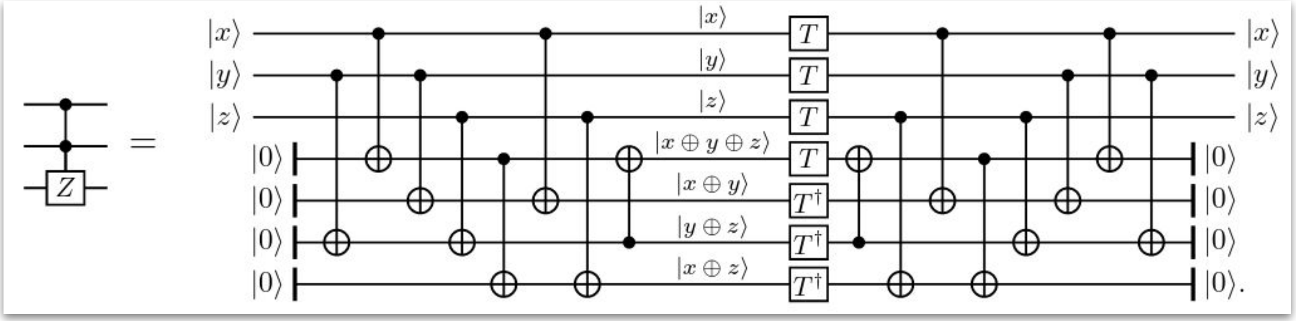
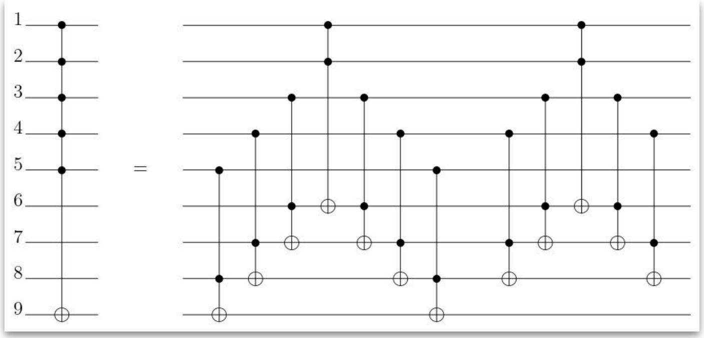


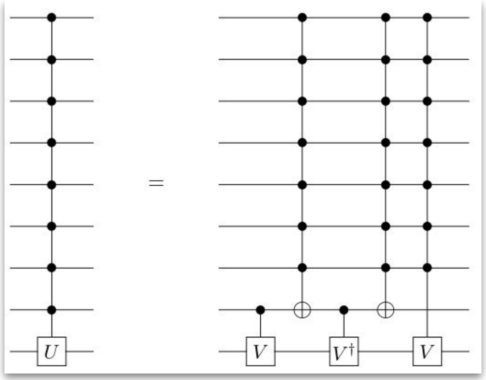
Image: P. Selinger (2013) *Quantum circuits of T -depth one*. Phys. Rev. A 87, 042302 (2013)

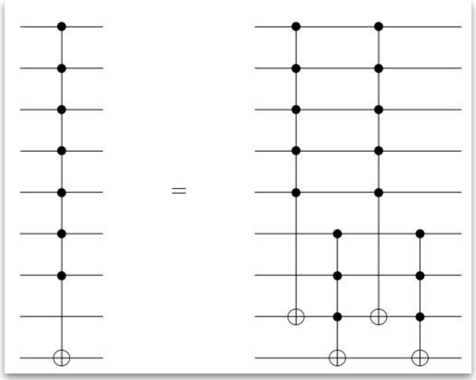
Example

Let’s decompose a multi-controlled NOT.

Multi-controlled NOT decomposition strategies







Images: Barenco et al., *Elementary gates for quantum computation* Phys.Rev. A 52 (1995) 3457

Next time

Content:

- The quantum compilation stack (from top to bottom)
- Quantum circuit optimization
- Hands-on: compilation and resource estimation of Grover's algorithm

Action items:

- Project group and paper selection due Thursday

Recommended reading:

- Nielsen and Chuang 4.5
- Compiling quantum circuits (PennyLane documentation)
- Explore the **qml.transforms** module